

Jay Desai

C.N. Yang Institute of Theoretical Physics,
Stony Brook, NY 11794, USA
✉ jay.desai@stonybrook.edu
Stony Brook University

Current Research Interests

- **Standard Model Effective Field Theory**
Model-independent search for hints of New Physics. Universal theories of SMEFT.
- **High Energy Jets Phenomenology**
Understanding the properties of Higgs jets and QCD jet substructure properties in collider phenomenology.

Education

- 2020-ongoing **Doctor of Philosophy**, *Stony Brook University*.
Physics. Advisor: M.C. Gonzalez-Garcia, George Sterman
- 2018-2020 **Master of Arts**, *Stony Brook University*.
Physics
- 2014-2018 **Bachelor of Technology**, *Indian Institute of Technology Roorkee*.
Electronics and Communication Engineering with Minor Specialization in Physics

Publications

All papers can be found on my [INSPIRE](#) and [Google Scholar](#) profile

- “Drell-Yan production in universal theories beyond dimension-six SMEFT”
[arXiv:2503.19962 \[hep-ph\]](#), *Phys. Rev. D* **112**, 013009 (2025)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz
- “Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
[arXiv:2404.03720 \[hep-ph\]](#), *Phys. Rev. D* **110** (2024) 033003
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- “Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT” [arXiv:2304.03305 \[hep-ph\]](#), *Phys. Rev. D* **107**, 115013 (2023)
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz

In preparation

- *A phenomenological study of Higgs jets at a muon collider* (To appear [arXiv:26XX](#))
Collaborators: George Sterman, Patrick Meade
- *Constraining universal extensions of the Standard Model beyond dimension-six SMEFT*
(To appear [arXiv:26XX](#))
Collaborators: Tyler Corbett, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martines

Past academic projects

- *Understanding the contribution of the strange quark sea to the spin of the proton* (Master's Thesis)
Advisor: Abhay Deshpande

Seminars/Conference Talks

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
"Universal Standard Model Effective Field Theory beyond dimension-six"
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
"Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **LHC EFT WG meeting, (online), April 28th 2025**
"Drell-Yan production in universal theories beyond dimension-six SMEFT"
- **DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
"Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **Pheno 2023, University of Pittsburgh, May 8th 2023**
"A Phenomenological study of Higgs jets a muon collider"
- **Higgs 2022, Pisa, Italy, Nov 9th 2022(hybrid)**
"A Phenomenological study of Higgs jets a muon collider"
- **EIC Yellow Report: semi-inclusive working group meeting**
"Impact study of Kaon SIDIS at EIC on polarized strange quark distribution function"
- **APS April Meeting, 2020**
"Strange quark polarized distribution function using SIDIS at EIC"
- **YITP Pheno Meeting, September 2025**
"Universal Standard Model Effective Field Theory beyond dimension-six"
- **YITP Pheno Meeting, February 2024**
"Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"

Awards and Fellowships

- Awarded the **Gerald Brown Prize**, a research travel award at Stony Brook University to attend **Pheno '23** conference.
- Attended **"Invisibles24"** Summer School and Workshop at the University of Bologna, Italy from June 24th to July 5th, 2024 supported by the **HIDDeN** network.
- **Visiting researcher** at the *Departament de Física Quàntica i Astrofísica* of the *Facultat de Física de la Universitat de Barcelona* from Nov 1st to Nov 30th, 2023.
- Awarded the **NEST-Nationwide Education and Scholarship Test**, Junior Scholarship 2013 and stood 13th in the country.

Academic service

- Journal Referee, **JHEP**.

Teaching Experience

- PHY 133: Classical Physics Laboratory I, Instructor, Fall 2020
- PHY 134: Classical Physics Laboratory II, Instructor, Spring 2021
- PHY 301: Electromagnetic Theory I, TA, Fall 2021
- PHY 308: Quantum Physics, TA, Spring 2022

Skills

Programming Languages Python, C, C++, Java

Software MATLAB, Mathematica, FeynRules, FeynCalc, MATCHETE, MadGraph5, Pythia, ROOT,
Packages DJANGO